LOCAL SCALE-INVARIANT CONCENTRATION AND THE TYPE I/TYPE II DICHOTOMY FOR NAVIER–STOKES

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Abstract. We prove a local analytic reduction for possible finite-time singularities of the three-dimensional Navier–Stokes equations. For a suitable weak solution, every singular point at time T has arbitrarily small backward parabolic cylinders on which both the standard mean-subtracted Caffarelli–Kohn–Nirenberg scale-invariant quantity $C + D$ and the scale-invariant velocity quantity C are bounded below by positive constants. If these local scale-invariant quantities tend to zero at each point of the singular set, the solution is regular at time T by the ε-regularity theorem. Along any such positive concentration sequence, the solution either satisfies a local Type I scale-invariant bound or falls under the local Type II alternative. For local pointwise Type I terminal singularities of suitable weak solutions, the endpoint weak-Serrin bridge in the companion residual paper gives the velocity no-escape condition used in the Type I blow-up analysis.

1. Introduction

Let (u, p) be a suitable weak solution of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + u \cdot \nabla u + \nabla p = \Delta u, \qquad \nabla \cdot u = 0.$$

We use the standard normalization in which the viscosity is one. The weak solution theory goes back to Leray [6]. Suitable weak solutions and their partial regularity theory were introduced and developed by Scheffer [8] and by Caffarelli, Kohn, and Nirenberg [2]; see also Lin [7]. This paper establishes a local reduction from a possible singularity at time T to the Type I/Type II blow-up rate dichotomy.

Scale-invariant regularity and blow-up criteria provide the standard background for this formulation; see Serrin [10], Giga [5], Escauriaza–Seregin–Šverak [3], and Seregin [9]. The principal regularity input used below is the Caffarelli–Kohn–Nirenberg ε-regularity theorem.

The proof proceeds in four steps. First, the local scale-invariant quantities C and D are finite for suitable weak solutions and invariant under the usual parabolic scaling. Second, the Caffarelli–Kohn–Nirenberg ε-regularity theorem implies that vanishing of $C + D$ at a candidate singular point gives local regularity, while the velocity ε-regularity criterion gives positive concentration in the scale-invariant $L^3_{x,t}$ quantity. Third, a finite-time singularity therefore produces sequences of shrinking cylinders on which $C + D$, and separately C, are bounded below by positive constants. Finally, once such a sequence is fixed, it either satisfies a Type I scale-invariant bound or falls under the local Type II alternative.

The concentration and Type I/Type II dichotomy arguments are local in spacetime. Apart from the standard Caffarelli–Kohn–Nirenberg ε-regularity theorem, all quantities and alternatives used in that local argument are defined below. The terminal bridge section records how the companion residual paper upgrades the local pointwise Type I envelope to terminal velocity no-escape through the endpoint weak-Serrin local Type I package.

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2. Suitable Weak Solutions and Singular Points

Definition 2.1 (Suitable weak solution)**.** Let $I \subset \mathbb{R}$ be an open interval. A pair (u,p) is a suitable weak solution on $\mathbb{R}^3 \times I$ if

$$u \in L_t^\infty L_{x,\mathrm{loc}}^2 \cap L_t^2 H_{x,\mathrm{loc}}^1, \qquad p \in L_{\mathrm{loc}}^{3/2}, \qquad \nabla \cdot u = 0,$$

the Navier–Stokes equations hold in the sense of distributions, and the local energy inequality holds: for almost every $t_1 < t_2$ in I and every nonnegative $\phi \in C_c^\infty(\mathbb{R}^3 \times I)$,

$$\begin{aligned}
&\int_{\mathbb{R}^3} \frac{|u(x,t_2)|^2}{2}\phi(x,t_2)\,dx + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} |\nabla u|^2 \phi\,dx\,dt \\
&\le \int_{\mathbb{R}^3} \frac{|u(x,t_1)|^2}{2}\phi(x,t_1)\,dx + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} \frac{|u|^2}{2}(\partial_t\phi + \Delta\phi)\,dx\,dt \\
&\quad + \int_{t_1}^{t_2}\int_{\mathbb{R}^3} \left(\frac{|u|^2}{2} + p\right) u\cdot\nabla\phi\,dx\,dt.
\end{aligned}$$

The pressure is understood modulo functions of time.

For $z_0 = (x_0, t_0)$ and $r > 0$, write

$$Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0).$$

We also write $Q_r(x_0,t_0)$ for $Q_r((x_0,t_0))$.

Definition 2.2 (Singular set at a fixed time)**.** For a time T, define

$$\Sigma(T) = \{x \in \mathbb{R}^3 :\ u \text{ is not locally bounded in any } Q_r(x,T)\}.$$

Equivalently, $x \notin \Sigma(T)$ if there are $r > 0$ and $M < \infty$ such that $\|u\|_{L^\infty(Q_r(x,T))} \le M$.

If a finite-time singularity occurs at time T, then $\Sigma(T) \neq \emptyset$ in this local sense.

3. Parabolic Rescaling and Scale-Invariant Quantities

Fix $z_0 = (x_0, T)$. For $r > 0$, define the parabolic rescaling

$$u^{(r)}(y,s) = r\,u(x_0 + ry, T + r^2 s), \qquad p^{(r)}(y,s) = r^2 p(x_0 + ry, T + r^2 s).$$

The rescaled variables are defined on the set of (y,s) for which $(x_0 + ry, T + r^2 s)$ belongs to the original spacetime domain.

Proposition 3.1 (Invariance under parabolic rescaling)**.** *If (u,p) is a suitable weak solution near (x_0,T), then $(u^{(r)}, p^{(r)})$ is a suitable weak solution on the rescaled domain. The local energy inequality is preserved, and the pressure remains defined modulo functions of the rescaled time.*

Proof. The distributional equations and the divergence condition follow from the change of variables $x = x_0 + ry$, $t = T + r^2 s$. We include the verification of the local energy inequality to fix the normalization.

Let $\psi \in C_c^\infty$ be nonnegative and compactly supported in the rescaled domain. Define

$$\phi(x,t) = \psi\left(\frac{x - x_0}{r}, \frac{t - T}{r^2}\right).$$

If $y = (x - x_0)/r$ and $s = (t - T)/r^2$, then

$$\partial_t\phi = r^{-2}(\partial_s\psi)(y,s), \qquad \nabla_x\phi = r^{-1}(\nabla_y\psi)(y,s), \qquad \Delta_x\phi = r^{-2}(\Delta_y\psi)(y,s).$$

For almost every $s_1 < s_2$, the corresponding times $T + r^2 s_1 < T + r^2 s_2$ are admissible in the original local energy inequality. Apply that inequality for (u, p) to ϕ on $[T + r^2 s_1, T + r^2 s_2]$. After the change of variables $x = x_0 + ry$, $t = T + r^2 s$, using

$$u(x_0 + ry, T + r^2 s) = r^{-1} u^{(r)}(y, s), \qquad p(x_0 + ry, T + r^2 s) = r^{-2} p^{(r)}(y, s),$$

each term contains the same common factor r. Dividing by this factor gives the rescaled inequality

$$\begin{aligned}&\int_{\mathbb{R}^3} \frac{|u^{(r)}(y, s_2)|^2}{2} \psi(y, s_2)\, dy + \int_{s_1}^{s_2} \int_{\mathbb{R}^3} |\nabla_y u^{(r)}|^2 \psi \, dy\, ds \\ &\leq \int_{\mathbb{R}^3} \frac{|u^{(r)}(y, s_1)|^2}{2} \psi(y, s_1)\, dy + \int_{s_1}^{s_2} \int_{\mathbb{R}^3} \frac{|u^{(r)}|^2}{2} (\partial_s \psi + \Delta_y \psi)\, dy\, ds \\ &\quad + \int_{s_1}^{s_2} \int_{\mathbb{R}^3} \left(\frac{|u^{(r)}|^2}{2} + p^{(r)} \right) u^{(r)} \cdot \nabla_y \psi \, dy\, ds.\end{aligned}$$

Thus $(u^{(r)}, p^{(r)})$ is suitable on the rescaled domain. The pressure transforms according to the Navier–Stokes scaling, and addition of a function of t becomes addition of a function of s. $\square$

Definition 3.2 (Scale-invariant local quantities)**.** For $z_0 = (x_0, t_0)$ and $r > 0$, set

$$C(z_0, r) = r^{-2} \int_{Q_r(z_0)} |u|^3 \, dx\, dt$$

and

$$D(z_0, r) = r^{-2} \int_{t_0 - r^2}^{t_0} \int_{B_r(x_0)} |p(x, t) - (p)_{B_r(x_0)}(t)|^{3/2} \, dx\, dt.$$

Here $(p)_{B_r(x_0)}(t)$ denotes the spatial mean of $p(\cdot, t)$ on $B_r(x_0)$.

The subtraction of the spatial mean removes the pressure ambiguity in the form used by the Caffarelli–Kohn–Nirenberg criterion.

Proposition 3.3 (Local finiteness and scaling)**.** *Let (u, p) be a suitable weak solution near $z_0 = (x_0, T)$. Then $C(z_0, r)$ and $D(z_0, r)$ are finite for every sufficiently small r. Moreover, for every fixed $R < \infty$,*

$$\begin{aligned}&\int_{-R^2}^{0} \int_{B_R} \left(|u^{(r)}|^3 + |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2} \right) dy\, ds \\ &\qquad = R^2 \big(C(z_0, rR) + D(z_0, rR) \big).\end{aligned}$$

Proof. Local finiteness of the velocity term follows from

$$u \in L^\infty_t L^2_{x,\mathrm{loc}} \cap L^2_t H^1_{x,\mathrm{loc}} \subset L^3_{\mathrm{loc}}$$

on finite cylinders. Indeed, if $B \Subset \mathbb{R}^3$ and $(a, b) \Subset I$, Sobolev and interpolation give

$$\int_a^b \|u(t)\|_{L^3(B)}^3 \, dt \leq C \left(\operatorname*{ess\,sup}_{a<t<b} \|u(t)\|_{L^2(B)} \right)^{3/2} \left(\int_a^b \|u(t)\|_{H^1(B)}^2 \, dt \right)^{3/4} (b - a)^{1/4}.$$

Thus $u \in L^3_{\mathrm{loc}}$. The pressure term is finite because $p \in L^{3/2}_{\mathrm{loc}}$, and subtracting a spatial mean over the same ball preserves local $L^{3/2}$ integrability. More explicitly, for a ball B,

$$|p - (p)_B(t)|^{3/2} \leq C \left(|p|^{3/2} + |(p)_B(t)|^{3/2} \right),$$

while Jensen's inequality gives

$$|(p)_B(t)|^{3/2} \leq \frac{1}{|B|} \int_B |p(x, t)|^{3/2} \, dx.$$

After integration over B and over time, the mean-subtracted pressure term is controlled by the local $L^{3/2}$ norm of p.

For the scaling identity, use $x = x_0 + ry$, $t = T + r^2 s$. The velocity term transforms as

$$\int_{-R^2}^{0} \int_{B_R} |u^{(r)}|^3 \, dy \, ds = r^{-2} \int_{T-(rR)^2}^{T} \int_{B_{rR}(x_0)} |u|^3 \, dx \, dt.$$

The pressure means transform according to

$$(p^{(r)})_{B_R}(s) = r^2 (p)_{B_{rR}(x_0)}(T + r^2 s),$$

because

$$\frac{1}{|B_R|} \int_{B_R} r^2 p(x_0 + ry, T + r^2 s) \, dy = \frac{r^2}{|B_{rR}|} \int_{B_{rR}(x_0)} p(x, T + r^2 s) \, dx.$$

Therefore

$$\begin{aligned} &\int_{-R^2}^{0} \int_{B_R} |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2} \, dy \, ds \\ &\qquad = r^{-2} \int_{T-(rR)^2}^{T} \int_{B_{rR}(x_0)} |p - (p)_{B_{rR}(x_0)}(t)|^{3/2} \, dx \, dt. \end{aligned}$$

Adding the velocity and pressure identities gives the displayed formula. ◻

4. ε-Regularity and the Vanishing Case

Theorem 4.1 (Caffarelli–Kohn–Nirenberg ε-regularity)**.** *There exists a universal constant $\varepsilon_0 > 0$ with the following property. Let (u, p) be a suitable weak solution in $Q_r(z_0)$. If*

$$C(z_0, r) + D(z_0, r) < \varepsilon_0,$$

then u is locally bounded in a smaller cylinder, for instance in $Q_{r/2}(z_0)$. In particular, if $z_0 = (x_0, T)$, then $x_0 \notin \Sigma(T)$.

Proof. This is the local ε-regularity theorem of Caffarelli–Kohn–Nirenberg [2]; see also Lin [7]. It is stated here with the spatial mean subtraction used in D. Equivalent representatives of the pressure differ by functions of time, and the spatial mean subtraction on $B_r(x_0)$ removes this ambiguity. ◻

Theorem 4.2 (Velocity ε-regularity)**.** *There exists a universal constant $\varepsilon_v > 0$ with the following property. Let (u, p) be a suitable weak solution in $Q_r(z_0)$. If*

$$C(z_0, r) < \varepsilon_v,$$

then u is locally bounded in $Q_{r/2}(z_0)$. In particular, if $z_0 = (x_0, T)$, then $x_0 \notin \Sigma(T)$.

Proof. This is the standard $L^3_{x,t}$ form of the Caffarelli–Kohn–Nirenberg ε-regularity criterion. It follows from the standard local pressure decay lemma together with Theorem 4.1; equivalently, it is the $L^3_{x,t}$-smallness criterion proved in the Caffarelli–Kohn–Nirenberg theory and in Lin's direct proof [2, 7]. The statement is scale invariant because C is invariant under the parabolic Navier–Stokes scaling. ◻

Definition 4.3 (Vanishing local scale-invariant quantities at time T)**.** We say that the local scale-invariant quantities vanish at time T if, for every $x_0 \in \Sigma(T)$,

$$\limsup_{r \downarrow 0} \big(C((x_0, T), r) + D((x_0, T), r) \big) = 0.$$

This condition is vacuous when $\Sigma(T) = \emptyset$.

Proposition 4.4 (Equivalent rescaled formulation)**.** *The preceding condition is equivalent to the following statement: for every $x_0 \in \Sigma(T)$ and every fixed $R < \infty$,*

$$\lim_{r \downarrow 0} \int_{-R^2}^{0} \int_{B_R} \left(|u^{(r)}|^3 + |p^{(r)} - (p^{(r)})_{B_R}(s)|^{3/2} \right) dy\, ds = 0.$$

Proof. By Proposition 3.3, the rescaled integral over $B_R \times (-R^2, 0)$ equals

$$R^2 \big(C((x_0,T), rR) + D((x_0,T), rR) \big).$$

Thus vanishing of $C + D$ implies vanishing on every fixed rescaled cylinder. Conversely, taking $R = 1$ gives the original pointwise condition. □

Theorem 4.5 (Regularity under vanishing local quantities)**.** *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$ for some $\delta > 0$. If the local scale-invariant quantities vanish at time T, then $\Sigma(T) = \emptyset$. Thus no finite-time singularity occurs at time T in the no-concentration case.*

Proof. Suppose, toward a contradiction, that $x_0 \in \Sigma(T)$. By the vanishing hypothesis, choose $r > 0$ such that

$$C((x_0,T), r) + D((x_0,T), r) < \varepsilon_0.$$

Theorem 4.1 implies that u is bounded in $Q_{r/2}(x_0, T)$, hence $x_0 \notin \Sigma(T)$. This contradicts the choice of x_0. Hence $\Sigma(T) = \emptyset$. □

Corollary 4.6 (Vanishing local quantities exclude a terminal singularity)**.** *Let (u, p) be suitable near (x_0, T). If*

$$\limsup_{r \downarrow 0} \big(C((x_0,T), r) + D((x_0,T), r) \big) = 0,$$

then u is bounded in some backward cylinder $Q_\rho(x_0, T)$, and hence $x_0 \notin \Sigma(T)$. Equivalently, membership in $\Sigma(T)$ requires that $C + D$ have a positive lower bound along a sequence of shrinking backward parabolic cylinders.

Proof. The hypothesis gives a scale at which $C + D$ is below the universal threshold in Theorem 4.1. The ε-regularity theorem gives local boundedness in a backward cylinder ending at T, so $x_0 \notin \Sigma(T)$. □

5. Positive Local Concentration

Lemma 5.1 (Failure of vanishing gives a positive concentration sequence)**.** *Suppose the local scale-invariant quantities do not vanish at time T. Then there exist $x_0 \in \Sigma(T)$, a constant $\eta > 0$, and radii $r_n \downarrow 0$ such that*

$$C((x_0,T), r_n) + D((x_0,T), r_n) \geq \eta \qquad \text{for all } n.$$

Equivalently, the rescaled sequence around (x_0, T) has a positive lower bound for the corresponding local scale-invariant quantities on a fixed bounded cylinder.

Proof. Negating the definition of vanishing gives a point $x_0 \in \Sigma(T)$ with positive limsup of

$$C((x_0,T), r) + D((x_0,T), r)$$

as $r \downarrow 0$. Choose $\eta > 0$ below that limsup and choose a sequence $r_n \downarrow 0$ along which the displayed lower bound holds. The rescaled formulation follows from Proposition 3.3. □

Definition 5.2 (Positive local concentration sequence)**.** Let $x_0 \in \Sigma(T)$. A sequence $r_n \downarrow 0$ is a positive local concentration sequence at (x_0, T) if there is $\eta > 0$ such that

$$C((x_0,T), r_n) + D((x_0,T), r_n) \geq \eta \qquad \text{for all } n.$$

The corresponding parabolic rescalings are those defined by

$$u_n(y,s) = r_n u(x_0 + r_n y, T + r_n^2 s), \qquad p_n(y,s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Proposition 5.3 (Persistence of positive concentration)**.** *Let (u_n, p_n) be suitable weak solutions on a fixed cylinder $B_R \times (-R^2, 0)$. Suppose that*

$$\int_{-R^2}^{0} \int_{B_R} \left(|u_n|^3 + |p_n - (p_n)_{B_R}(s)|^{3/2} \right) dy\, ds \geq \eta > 0$$

for all n. Suppose further that, after passing to a subsequence,

$$u_n \to U \quad \text{in } L^3(B_R \times (-R^2, 0))$$

and

$$p_n - (p_n)_{B_R}(s) \to \Pi \quad \text{in } L^{3/2}(B_R \times (-R^2, 0)).$$

Then

$$\int_{-R^2}^{0} \int_{B_R} \left(|U|^3 + |\Pi|^{3/2} \right) dy\, ds \geq \eta.$$

In particular, the limiting pair retains the same positive lower bound for the mean-subtracted local scale-invariant quantity. If $\Pi = P - (P)_{B_R}(s)$, this is the corresponding Caffarelli–Kohn–Nirenberg quantity of (U, P) on $B_R \times (-R^2, 0)$.

Proof. Strong convergence in L^3 gives

$$\int_{-R^2}^{0} \int_{B_R} |u_n|^3\, dy\, ds \longrightarrow \int_{-R^2}^{0} \int_{B_R} |U|^3\, dy\, ds.$$

Strong convergence in $L^{3/2}$ gives the analogous convergence of the mean-subtracted pressure integrals. Passing to the limit in the lower bound yields the asserted inequality. □

Remark 5.4. Proposition 5.3 is an auxiliary compactness observation, independent of the proof of the dichotomy below.

Proposition 5.5 (Local alternative)**.** *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$ for some $\delta > 0$. Exactly one of the following alternatives holds.*

(i) *For every $x_0 \in \Sigma(T)$,*

$$\limsup_{r \downarrow 0} \big(C((x_0, T), r) + D((x_0, T), r) \big) = 0.$$

(ii) *There exist $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \qquad \text{for all } n.$$

Proof. The first alternative is a universal assertion over the singular set, and the second is its failure expressed by a sequence. If the first alternative fails, then for some $x_0 \in \Sigma(T)$ the corresponding limsup is positive; choosing a positive number below this limsup and a sequence along which the lower bound holds gives the second alternative. Conversely, the second alternative contradicts the first. Hence exactly one alternative holds. □

Theorem 5.6 (Finite-time singularity yields positive local scale-invariant concentration)**.** *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$. If $\Sigma(T) \neq \emptyset$, then there are $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \qquad \text{for all } n.$$

Proof. If the first alternative in Proposition 5.5 held, then Theorem 4.5 would imply $\Sigma(T) = \emptyset$, contrary to the hypothesis. Therefore the second alternative in Proposition 5.5 holds. □

Corollary 5.7 (Singular points carry positive velocity concentration). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$. If $x_0 \in \Sigma(T)$, then there are $\eta_v > 0$ and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) \geq \eta_v \qquad \textit{for all } n.$$

Proof. If the conclusion failed, then

$$\limsup_{r \downarrow 0} C((x_0, T), r) = 0.$$

Choose $r > 0$ so small that

$$C((x_0, T), r) < \varepsilon_v,$$

where ε_v is the constant in Theorem 4.2. The velocity ε-regularity theorem gives boundedness of u in $Q_{r/2}(x_0, T)$, contradicting $x_0 \in \Sigma(T)$. Hence the limsup is positive; taking any η_v below it and passing to a sequence with $C((x_0, T), r_n) \geq \eta_v$ gives the result. □

6. Automatic No-Escape for Finite-Energy Type I Singularities

The preceding concentration result gives the positive velocity concentration sequence used in the Type I blow-up analysis. In the companion paper, the remaining compactness input is a no-escape condition excluding the possibility that all of the selected scale-invariant velocity mass collapses into the final rescaled time slice. We prove here that this no-escape condition is automatic for finite-energy Type I singularities.

For this section we use, in addition to C and D, the standard local kinetic-energy and dissipation quantities

$$A(z_0, r) = r^{-1} \operatorname*{ess\,sup}_{t_0 - r^2 < t < t_0} \int_{B_r(x_0)} |u(x,t)|^2 \, dx,$$

and

$$E(z_0, r) = r^{-1} \int_{t_0 - r^2}^{t_0} \int_{B_r(x_0)} |\nabla u(x,t)|^2 \, dx \, dt.$$

These quantities are invariant under the Navier–Stokes scaling.

Definition 6.1 (Finite-energy Type I terminal singularity). Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (0, T)$. We say that $z_* = (x_*, T)$ is a finite-energy Type I terminal singularity if $x_* \in \Sigma(T)$,

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; \dot{H}^1(\mathbb{R}^3)),$$

and there are $T_0 < T$ and $M < \infty$ such that

$$\|u(t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{M}{\sqrt{T - t}}, \qquad T_0 < t < T. \tag{1}$$

The key point is a uniformly-local energy propagation estimate. It is stated at unit scale, because the application is obtained by parabolic rescaling from (x_*, T). The estimate is a local state-space closure: the state is the uniformly-local kinetic energy, and the Type I envelope turns the nonlinear pressure stratum into an integrable linear coefficient. No bound below depends on the size of the global L^2 energy or the global dissipation; finite energy is used only to place the pressure in the standard whole-space Leray gauge and to make fixed-scale terminal L^3 absolute continuity immediate.

We use the standard Leray pressure gauge throughout this section. At almost every terminal time on which the Type I bound holds, $u(t) \in L^2(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$, hence $u_i(t) u_j(t) \in L^{3/2}(\mathbb{R}^3)$. The whole-space pressure

$$p^L(t) = \mathcal{R}_i \mathcal{R}_j (u_i(t) u_j(t))$$

is then defined by the Calderon–Zygmund theory. The distributional pressure equation gives $-\Delta p = \partial_i\partial_j(u_iu_j) = -\Delta p^L$, and the whole-space de Rham argument applied to the momentum equation shows that $p - p^L$ is a function of time. Thus replacing p by p^L is only a pressure-gauge change and does not alter suitability. This gauge choice uses the finite-energy setting to define the representative; the uniform constants below, in particular B_{uloc} and B_A, do not contain $\|u\|_{L^\infty(0,T;L^2(\mathbb{R}^3))}$ or $\int_0^T \|\nabla u(t)\|^2_{L^2(\mathbb{R}^3)}\,dt$.

Lemma 6.2 (Uniformly-local energy propagation under a Type I envelope)**.** *Let (v,π) be a suitable weak solution on $\mathbb{R}^3\times(-4,0)$ with $v(s)\in L^2(\mathbb{R}^3)$ for a.e. s, and take π to be the whole-space Leray pressure representative, modulo functions of time. Assume that*

$$|v(x,s)| \le M(-s)^{-1/2}, \qquad \text{for a.e. } (x,s)\in\mathbb{R}^3\times(-4,0), \tag{2}$$

and that for some admissible initial time $s_0\in(-4,-3)$,

$$\alpha_0 := \sup_{a\in\mathbb{R}^3}\int_{B_2(a)} |v(x,s_0)|^2\,dx \le K. \tag{3}$$

Then there is a constant $B_{\mathrm{uloc}} = B_{\mathrm{uloc}}(M,K) < \infty$ such that

$$\operatorname*{ess\,sup}_{-1<s<0}\sup_{a\in\mathbb{R}^3}\int_{B_1(a)}|v(x,s)|^2\,dx + \sup_{a\in\mathbb{R}^3}\int_{-1}^0\int_{B_1(a)}|\nabla v|^2\,dx\,ds \le B_{\mathrm{uloc}}. \tag{4}$$

Proof. We give the proof because the estimate is the mechanism that closes the terminal no-escape gap. Let

$$\alpha(s) = \sup_{a\in\mathbb{R}^3}\int_{B_1(a)}|v(x,s)|^2\,dx.$$

By (2),

$$\alpha(s)^{1/2} \le CM(-s)^{-1/2}, \qquad -4<s<0. \tag{5}$$

Fix $a\in\mathbb{R}^3$ and choose $\phi_a\in C_c^\infty(B_2(a))$ with $0\le\phi_a\le1$, $\phi_a\equiv1$ on $B_1(a)$, and $|\nabla\phi_a|+|\Delta\phi_a|\le C$, uniformly in a. Applying the local energy inequality on (s_0,t), using the standard monotone approximation of temporal cutoffs and with the pressure gauge chosen below, gives for a.e. $t\in(s_0,0)$

$$\begin{aligned}
&\int |v(t)|^2\phi_a\,dx + 2\int_{s_0}^t\int|\nabla v|^2\phi_a\,dx\,ds\\
&\quad\le \int|v(s_0)|^2\phi_a\,dx + C\int_{s_0}^t\int_{B_2(a)}|v|^2\,dx\,ds + C\int_{s_0}^t\int_{B_2(a)}|v|^3\,dx\,ds\\
&\qquad + C\int_{s_0}^t\int_{B_2(a)}|\pi - c_a(s)|\,|v|\,dx\,ds.
\end{aligned}\tag{6}$$

Here $c_a(s)$ is an arbitrary function of time; adding or subtracting such a function does not change the local energy inequality, since $\nabla\cdot v=0$ and ϕ_a is compactly supported.

The velocity terms are controlled by the state. Since $B_2(a)$ is covered by finitely many unit balls,

$$\int_{B_2(a)}|v|^2\,dx \le C\alpha(s),$$

and by (2),

$$\int_{B_2(a)}|v|^3\,dx \le M(-s)^{-1/2}\int_{B_2(a)}|v|^2\,dx \le CM(-s)^{-1/2}\alpha(s). \tag{7}$$

It remains to estimate the pressure term uniformly in a.

Use the whole-space pressure representative $\pi = \mathcal{R}_i\mathcal{R}_j(v_iv_j)$, modulo functions of time. Let $\chi_a \in C_c^\infty(B_8(a))$, $\chi_a \equiv 1$ on $B_4(a)$, and write inside $B_4(a)$

$$\pi = q_a + h_a, \qquad q_a = \mathcal{R}_i\mathcal{R}_j(\chi_a v_i v_j).$$

Then h_a is harmonic in $B_4(a)$. Calderon–Zygmund boundedness [11] and (2) give, for a.e. s,

$$\begin{aligned}\|q_a(s)\|_{L^2(B_4(a))} &\le C\|\chi_a v(s)\otimes v(s)\|_{L^2(\mathbb{R}^3)}\\ &\le CM(-s)^{-1/2}\|v(s)\|_{L^2(B_8(a))}\\ &\le CM(-s)^{-1/2}\alpha(s)^{1/2},\end{aligned}$$

where the last step again uses a finite covering by unit balls. Hence

$$\int_{B_2(a)} |q_a|\,|v|\,dx \le CM(-s)^{-1/2}\alpha(s). \tag{8}$$

For the harmonic part, set $c_a(s) = h_a(a,s)$. The kernel estimate

$$|K_{ij}(x-z) - K_{ij}(a-z)| \le C|x-a|\,|z-a|^{-4}, \qquad x\in B_2(a), \quad |z-a|>4,$$

where K_{ij} is the kernel of $\mathcal{R}_i\mathcal{R}_j$, gives

$$|h_a(x,s) - c_a(s)| \le C\sum_{m=2}^{\infty} 2^{-4m}\int_{2^m<|z-a|<2^{m+1}} |v(z,s)|^2\,dz.$$

The annulus $\{2^m < |z-a| < 2^{m+1}\}$ is covered by at most $C2^{3m}$ unit balls. Therefore

$$\|h_a(s) - c_a(s)\|_{L^\infty(B_2(a))} \le C\sum_{m=2}^{\infty} 2^{-m}\alpha(s) \le C\alpha(s). \tag{9}$$

Consequently,

$$\begin{aligned}\int_{B_2(a)} |h_a - c_a|\,|v|\,dx &\le C\alpha(s)\int_{B_2(a)}|v|\,dx\\ &\le C\alpha(s)^{3/2}\\ &\le CM(-s)^{-1/2}\alpha(s),\end{aligned} \tag{10}$$

where the last inequality is exactly the Type I state-space closure (5). Combining (8) and (10),

$$\int_{B_2(a)} |\pi - c_a(s)|\,|v|\,dx \le CM(-s)^{-1/2}\alpha(s). \tag{11}$$

Insert (7) and (11) into (6), then take the supremum over a. Since $(-s)^{-1/2}\in L^1(-4,0)$, Gronwall's inequality yields

$$\operatorname*{ess\,sup}_{s_0<s<0}\alpha(s) \le CK\exp\left(C\int_{s_0}^0 \left(1+M(-s)^{-1/2}\right)ds\right) \le B_1(M,K).$$

Returning to (6), using this bound, and then letting $t\uparrow 0$ through admissible times gives the corresponding uniformly-local dissipation estimate on $(-1,0)$. This proves (4). □

Remark 6.3 (State-space stratification). The preceding proof can be read as a three-stratum closure of the local state $\alpha(s)$. The diffusion and transport strata are linear in α, with coefficient $1+M(-s)^{-1/2}$. The local pressure stratum is also linear after Calderon–Zygmund and the Type I envelope. The only apparently superlinear stratum is the harmonic pressure flux, which is $O(\alpha^{3/2})$; the pointwise Type I envelope implies $\alpha^{1/2}\le CM(-s)^{-1/2}$, reducing it to the same integrable linear coefficient. Thus the bad region $\alpha \gg 1$ is not invariant: the state-space dynamics are trapped in a bounded absorbing interval depending only on M and the initial uniformly-local energy state.

Proposition 6.4 (Automatic terminal kinetic tightness)**.** *Let (u, p, z_*), $z_* = (x_*, T)$, be a finite-energy Type I terminal singularity in the sense of Definition 6.1. Then there exist $r_0 > 0$ and $B_A < \infty$, with B_A depending only on the Type I constant M, such that*

$$A(z_*, r) \le B_A, \qquad 0 < r < r_0. \tag{12}$$

Moreover,

$$r^{-1} \int_{T-r^2}^{T} \int_{B_r(x_*)} |\nabla u|^2 \, dx \, dt \le B_A, \qquad 0 < r < r_0. \tag{13}$$

Proof. Choose $r_0 > 0$ so small that $T - 4r^2 > T_0$ for $0 < r < r_0$. For such r, use the whole-space Leray pressure gauge fixed above and define the rescaled solution

$$v^{(r)}(y, s) = ru(x_* + ry, T + r^2 s), \qquad \pi^{(r)}(y, s) = r^2 p(x_* + ry, T + r^2 s),$$

on $\mathbb{R}^3 \times (-4, 0)$. The Type I bound (1) gives

$$|v^{(r)}(y, s)| \le M(-s)^{-1/2}, \qquad -4 < s < 0.$$

For any admissible $s_0 \in (-4, -3)$ at which the Type I bound holds, and any $a \in \mathbb{R}^3$,

$$\begin{aligned}\int_{B_2(a)} |v^{(r)}(y, s_0)|^2 \, dy &= r^{-1} \int_{B_{2r}(x_*+ra)} |u(x, T + r^2 s_0)|^2 \, dx \\ &\le r^{-1} |B_{2r}| \frac{M^2}{r^2 |s_0|} \le CM^2.\end{aligned}$$

Such times form a full-measure subset of $(-4, -3)$. The preceding estimate is uniform in this choice, so Lemma 6.2 applies with $K = CM^2$, uniformly in r. Hence

$$\operatorname*{ess\,sup}_{-1<s<0} \int_{B_1} |v^{(r)}(y, s)|^2 \, dy + \int_{-1}^{0} \int_{B_1} |\nabla v^{(r)}|^2 \, dy \, ds \le B_A.$$

Undoing the parabolic change of variables gives exactly (12) and (13). The constant B_A depends only on M; no global energy norm of u has been used in the estimate. ☐

Theorem 6.5 (Automatic velocity no-escape)**.** *Let (u, p, z_*), $z_* = (x_*, T)$, be a finite-energy Type I terminal singularity. Let $\lambda_k \downarrow 0$ be any sequence of scales, and set*

$$u_k(x, t) = \lambda_k u(x_* + \lambda_k x, T + \lambda_k^2 t).$$

Then

$$\lim_{\sigma \downarrow 0} \sup_k \int_{-\sigma}^{0} \int_{B_1} |u_k(x, t)|^3 \, dx \, dt = 0. \tag{14}$$

Proof. By Proposition 6.4, after discarding at most finitely many indices,

$$\operatorname*{ess\,sup}_{-1<t<0} \int_{B_1} |u_k(x, t)|^2 \, dx \le B_A.$$

The Type I bound gives, for the same tail of the sequence,

$$\|u_k(t)\|_{L^\infty(\mathbb{R}^3)} \le M(-t)^{-1/2}, \qquad -1 < t < 0.$$

Therefore, for $0 < \sigma < 1$,

$$\int_{-\sigma}^{0}\int_{B_1} |u_k|^3\,dx\,dt \le \int_{-\sigma}^{0} \|u_k(t)\|_{L^\infty(B_1)} \int_{B_1} |u_k(x,t)|^2\,dx\,dt$$
$$\le MB_A \int_{-\sigma}^{0} (-t)^{-1/2}\,dt = 2MB_A\sigma^{1/2}.$$

This bound is uniform over the tail. For each of the finitely many discarded indices k, choose $\sigma_k > 0$ so that $T + \lambda_k^2 t > T_0$ for $-\sigma_k < t < 0$. On this terminal subinterval the Type I envelope and the finite-energy assumption imply

$$\int_{-\sigma_k}^{0}\int_{B_1} |u_k|^3\,dx\,dt \le \int_{-\sigma_k}^{0} \|u_k(t)\|_{L^\infty(B_1)} \int_{B_1} |u_k(x,t)|^2\,dx\,dt < \infty.$$

Indeed, on $(-\sigma_k, 0)$,

$$\|u_k(t)\|_{L^\infty(B_1)} \le M(-t)^{-1/2}, \qquad \int_{B_1} |u_k(x,t)|^2\,dx \le \lambda_k^{-1}\|u(T+\lambda_k^2 t)\|_{L^2(\mathbb{R}^3)}^2,$$

and the right-hand side is finite for this fixed k by finite energy. Thus each discarded index has absolute continuity of its L^3-integral at the terminal slice. Taking the maximum over this finite set gives the same limit for the full supremum over k. This proves (14). □

Corollary 6.6 (Automatic admissibility of Type I concentration sequences)**.** *Let (u, p, z_*), $z_* = (x_*, T)$, be a finite-energy Type I terminal singularity. Then there exist $\eta_v > 0$ and $\lambda_k \downarrow 0$ such that*

$$C(z_*, \lambda_k) \ge \eta_v \qquad \text{for every } k,$$

and the corresponding rescaled velocities satisfy the no-escape condition (14). Hence the concentration and no-escape admissibility condition used in the companion Type I blow-up-limit paper is automatic for finite-energy Type I singularities.

Proof. Since $x_* \in \Sigma(T)$, Corollary 5.7 gives $\eta_v > 0$ and $\lambda_k \downarrow 0$ with $C(z_*, \lambda_k) \ge \eta_v$ for all k. The no-escape condition for this same sequence follows from Theorem 6.5. □

7. The Type I/Type II Blow-Up Rate Dichotomy

Let $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ be given by Theorem 5.6. Define

$$u_n(y,s) = r_n u(x_0 + r_n y, T + r_n^2 s), \qquad p_n(y,s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Then (u_n, p_n) are suitable weak solutions on expanding backward cylinders, and by Proposition 3.3 they retain a positive lower bound for the mean-subtracted local scale-invariant quantity on a fixed bounded cylinder.

Definition 7.1 (Type I concentration sequence)**.** Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$, and let $r_n \downarrow 0$ be a positive local concentration sequence at (x_0, T). The sequence is called Type I if the concentration point (x_0, T) satisfies a local Type I scale-invariant bound: there are $\rho > 0$ and $M < \infty$, with $\rho^2 < \delta$, such that

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\,\|u(t)\|_{L^\infty(B_\rho(x_0))} \le M.$$

This is the Type I alternative.

Definition 7.2 (Local Type II alternative)**.** Let (u,p) be a suitable weak solution on $\mathbb{R}^3\times(T-\delta,T)$, and let $r_n \downarrow 0$ be a positive local concentration sequence at (x_0,T). The sequence belongs to the local Type II alternative if the concentration point (x_0,T) satisfies no local Type I scale-invariant bound. Equivalently, for every $\rho>0$ with $\rho^2<\delta$,

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\,\|u(t)\|_{L^\infty(B_\rho(x_0))} = \infty.$$

Theorem 7.3 (Local blow-up rate dichotomy)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3\times(T-\delta,T)$, and suppose $\Sigma(T)\neq\emptyset$. Then there exist $x_0\in\Sigma(T)$, $\eta>0$, and $r_n\downarrow 0$ such that*

$$C((x_0,T),r_n)+D((x_0,T),r_n)\geq \eta \qquad \text{for all } n.$$

The sequence r_n is a positive local concentration sequence. The corresponding concentration regime satisfies either the local Type I bound or the local Type II alternative. Consequently, once the vanishing case has been excluded by ε-regularity, every finite-time singular case falls under exactly one of the two blow-up rate alternatives.

Proof. If the local scale-invariant quantities vanished at every point of $\Sigma(T)$, then Theorem 4.5 would give $\Sigma(T)=\emptyset$, contrary to the hypothesis. Therefore Theorem 5.6 gives $x_0\in\Sigma(T)$, $\eta>0$, and $r_n\downarrow 0$ with the stated lower bound. The rescaled sequence has a positive local scale-invariant lower bound by Proposition 3.3. By definition, either the Type I bound in Definition 7.1 holds, or the local Type II alternative in Definition 7.2 holds. □

Remark 7.4. This theorem is local. Its conclusion is the existence of a positive mean-subtracted scale-invariant concentration sequence and the dichotomy between a local Type I bound and failure of every such bound at the concentration point.

8. Local Pointwise Type I Bridge to the Terminal Concentration Package

We record the interface used by the companion residual paper [4]. The local pointwise Type I alternative in Definition 7.1 is strong enough, after rescaling at the singular point, to give compactness on every compact negative-time cylinder. The companion residual paper also records the endpoint weak-Serrin bridge: for suitable weak solutions the local pointwise Type I envelope belongs to $L^{2,\infty}_t L^\infty_x$, hence the Albritton–Barker weak Serrin local Type I package gives the scale-invariant A-bound on smaller cylinders and therefore terminal velocity no-escape. Thus positive local Type I concentration sequences are admissible for the Paper I Seregin extraction.

Throughout this section $z_*=(x_*,T)$, and

$$Q_1(0,0)=B_1(0)\times(-1,0).$$

Definition 8.1 (Admissible local Type I bridge sequence)**.** For a local pointwise Type I point $z_*=(x_*,T)$, a sequence $r_n\downarrow 0$ is an admissible local Type I bridge sequence if the rescaled velocities

$$u_n(x,t)=r_n u(x_*+r_n x, T+r_n^2 t)$$

satisfy, for some $\eta_v>0$,

$$\int_{Q_1(0,0)} |u_n|^3\,dx\,dt \geq \eta_v \qquad \text{for every } n,$$

and

$$\lim_{\sigma\downarrow 0}\sup_n \int_{-\sigma}^0\int_{B_1} |u_n|^3\,dx\,dt = 0.$$

Proposition 8.2 (Local pointwise Type I rescalings have terminal compactness)**.** *Let (u,p) be a suitable weak solution on $\mathbb{R}^3\times(0,T)$ satisfying the finite-energy bounds*

$$u\in L^\infty(0,T;L^2(\mathbb{R}^3))\cap L^2(0,T;\dot H^1(\mathbb{R}^3)).$$

Let $z_*=(x_*,T)$ be a singular point, and assume that z_* satisfies the local pointwise Type I bound: there exist $\rho>0$ and $M<\infty$ such that

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T}\sqrt{T-t}\,\|u(t)\|_{L^\infty(B_\rho(x_*))}\le M. \tag{15}$$

For any sequence $r_n\downarrow 0$, define

$$u_n(x,t)=r_nu(x_*+r_nx,T+r_n^2t),\qquad p_n(x,t)=r_n^2p(x_*+r_nx,T+r_n^2t). \tag{16}$$

Then the following hold.

(i) *If*

$$K=B_R\times[-S,-\sigma]\Subset\mathbb{R}^3\times(-\infty,0),\qquad R,S<\infty,\quad \sigma>0,$$

then, for all sufficiently large n,

$$\|u_n\|_{L^\infty(K)}\le M\sigma^{-1/2}. \tag{17}$$

(ii) *On each such compact cylinder K, after subtracting time-dependent local pressure gauges $a_{n,K}(t)$, the sequence satisfies uniform local Seregin bounds*

$$\|u_n\|_{L^\infty_tL^2_x(K)}+\|\nabla u_n\|_{L^2(K)}+\|p_n-a_{n,K}(t)\|_{L^{3/2}(K)}\le C_K. \tag{18}$$

Here C_K may depend on K,M,ρ, and on the finite Leray energy level of the original solution, but it does not depend on n and does not use a terminal no-escape assumption.

(iii) *After passing to a subsequence, $(u_n,p_n-a_{n,K})$ converges on every compact $K\Subset\mathbb{R}^3\times(-\infty,0)$ to an ancient suitable weak solution (U,Q), with*

$$u_n\to U\quad\text{strongly in } L^q(K),\qquad 1\le q<\infty,$$

$$p_n-a_{n,K}(t)\rightharpoonup Q\quad\text{weakly in } L^{3/2}(K),$$

and

$$|U(x,t)|\le M(-t)^{-1/2},\qquad t<0.$$

Proof. Fix $K=B_R\times[-S,-\sigma]\Subset\mathbb{R}^3\times(-\infty,0)$. For large n,

$$r_nR<\rho/2,\qquad r_n^2S<\rho^2/2.$$

Thus $x_*+r_nx\in B_\rho(x_*)$ and $T+r_n^2t\in(T-\rho^2,T)$ for a.e. $(x,t)\in K$. The local pointwise Type I bound (15) gives

$$|u_n(x,t)|=r_n|u(x_*+r_nx,T+r_n^2t)|\le r_nM(T-(T+r_n^2t))^{-1/2}=M(-t)^{-1/2}\le M\sigma^{-1/2}.$$

This proves (17). It also gives uniform local $L^\infty_tL^2_x$ and $L^3_{x,t}$ bounds on K.

It remains to explain the pressure and dissipation bounds. Use the whole-space Leray pressure representative, which is legitimate in the finite-energy class up to addition of functions of time. Choose $\chi_R\in C_c^\infty(B_{8R})$, $\chi_R\equiv1$ on B_{4R}, and write inside B_{4R}

$$p_n=q_n+h_n,\qquad q_n=\mathcal{R}_i\mathcal{R}_j(\chi_Ru_{n,i}u_{n,j}).$$

Calderon–Zygmund boundedness and the local L^3-bound for u_n give a uniform $L^{3/2}$-bound for q_n. For h_n, take the gauge $a_{n,K}(t)=h_n(0,t)$. The harmonic pressure kernel estimate gives

$$|h_n(x,t)-a_{n,K}(t)|\le C_R\sum_{m\ge2}2^{-4m}\int_{2^mR<|z|<2^{m+1}R}|u_n(z,t)|^2\,dz,\qquad x\in B_{2R}.$$

For annuli whose physical image remains inside $B_\rho(x_*)$, the local Type I bound controls the L^2-mass by

$$CM^2\sigma^{-1}(2^m R)^3,$$

and the resulting series is geometric. For the outer annuli, the finite Leray energy gives

$$\int_{\mathbb{R}^3} |u_n(z,t)|^2\,dz = r_n^{-1}\int_{\mathbb{R}^3} |u(y,T+r_n^2 t)|^2\,dy,$$

while the first outer annulus has radius comparable to ρ/r_n. Hence the tail of the kernel series is bounded by

$$C_{\rho,R}\,r_n^3\|u\|^2_{L^\infty(0,T;L^2)},$$

which is uniform in n. This proves the pressure-gauge part of (18).

Apply the local energy inequality for (u_n,p_n) with a cutoff equal to one on K and supported in a slightly larger compact cylinder. The local kinetic, cubic velocity, and pressure-flux terms are bounded by the estimates just proved, so $\|\nabla u_n\|_{L^2(K)} \le C_K$. The Navier–Stokes equation then bounds $\partial_t u_n$ locally in a negative Sobolev space. Aubin–Lions compactness and the pressure bound give the asserted Seregin subsequence convergence. Passing to the limit in the equations and local energy inequality gives an ancient suitable weak solution, and the pointwise Type I bound passes to the limit. □

Corollary 8.3 (Local pointwise Type I bridge sequences are admissible)**.** *Under the hypotheses of Proposition 8.2, let $r_n \downarrow 0$ be any sequence such that the rescaled velocities satisfy*

$$\int_{Q_1(0,0)} |u_n|^3\,dx\,dt \ge \eta_v > 0 \qquad \textit{for all } n.$$

Then r_n is an admissible local Type I bridge sequence in the sense of Definition 8.1.

Proof. The displayed positive concentration lower bound is the first requirement in Definition 8.1. The companion residual paper proves, using the Albritton–Barker weak Serrin local Type I estimate, that the local pointwise Type I envelope implies terminal velocity no-escape for every rescaled sequence [1, 4]. This is exactly the second requirement in Definition 8.1. □

Lemma 8.4 (Local Type I concentration enters the terminal dichotomy)**.** *Under the hypotheses of Proposition 8.2, let $r_n \downarrow 0$ be a velocity concentration sequence satisfying*

$$\int_{Q_1(0,0)} |u_n|^3\,dx\,dt \ge \eta_v > 0 \qquad \textit{for all } n.$$

Then r_n is admissible by Corollary 8.3. There are $\sigma_0 \in (0,1)$ and $\eta_1 > 0$ such that

$$\int_{-1}^{-\sigma_0}\int_{B_1} |u_n|^3\,dx\,dt \ge \eta_1 \qquad \textit{for all } n. \tag{19}$$

Consequently, after applying the terminal concentration package of the companion residual paper [4] with a threshold $\varepsilon_{\mathrm{sd}} \le \eta_1$, the retained compact concentration issued from the origin recentering has no loss of local compactness. The paired terminal decomposition produces an active retained profile locus plus a residual remainder. With realized critical-tail stratification and discharge proved in the companion paper, and with active path-space recurrence proved there, the companion package excludes mixed compact–diffuse remainders, compact–noncompact remainders, and infinite active descendant branches. Thus the retained compact concentration is reduced to a single retained concentrating profile or a finite separated family of retained concentrating profiles.

Proof. The no-escape condition in Definition 8.1 gives $\sigma_0 > 0$ such that the terminal layer $B_1 \times (-\sigma_0, 0)$ carries at most half of the fixed Q_1 velocity mass. This proves (19). The compactness part follows from Proposition 8.2 on the compact negative-time cylinder $B_1 \times [-1, -\sigma_0]$. Strong local L^3 convergence on this cylinder passes the positive mass to the ancient limit. The local CKN measure in the terminal package dominates the velocity contribution, so after a finite covering of $B_1 \times (-1, -\sigma_0)$ by unit terminal cylinders, one origin-frame recentering is $\varepsilon_{\mathrm{sd}}$-concentrating for a fixed $\varepsilon_{\mathrm{sd}} > 0$ depending on η_1. This is the only point at which terminal no-escape is used.

The terminal exhaustion theorem of the companion residual paper is paired: it records the active retained profile locus together with a residual remainder. Its diffuse-defect trichotomy and critical-tail compactification route compact core plus diffuse/noncompact remainders to the realized critical-tail rigidity theorem, while its active path-space recurrence rules out infinite active cascades. Therefore only a single retained profile or a finite separated retained family remains at this bridge stage. □

Theorem 8.5 (Local Type I entry into the full exclusion pipeline)**.** *Let (u, p) be a finite-energy suitable weak solution on $\mathbb{R}^3 \times (0, T)$, and let $z_* = (x_*, T)$ be a singular point satisfying the local pointwise Type I bound* (15)*. Then the rescalings at z_* generate a bounded centered Type I Seregin profile with retained local concentration, and the companion terminal package applies to that profile. In particular, no finite-energy local pointwise Type I singular point exists in the completed Papers 0–IV pipeline.*

Proof. Choose the positive Q_1 velocity concentration sequence supplied by the local concentration theorem. By Corollary 8.3, this sequence satisfies terminal no-escape and is admissible. By Lemma 8.4, the local Type I singularity generates retained compact concentration with local compactness. The compactness statement gives an ancient Type I limit satisfying $|U(x,t)| \leq M(-t)^{-1/2}$; after the centered change of variables this is a bounded centered Type I Seregin profile. If all retained profiles generated by the terminal package belong to lower classes already closed in Papers I–III, then the contradiction is supplied by those earlier class exclusions. If some retained profile lies outside the lower classes, it is a residual candidate. For residual candidates, the terminal local stratification theorem of the residual paper applies: finite separated retained families are excluded there, infinite active branches are excluded by active path-space recurrence, mixed compact–diffuse tails are routed through critical-tail compactification and realized critical-tail rigidity, and a remaining single retained profile is forced through indecomposability, parasitic-free mildness inheritance, and the endpoint Liouville contradiction. Thus the assumed local Type I singular point cannot exist in the completed pipeline. □

Corollary 8.6 (Local Type I alternatives are excluded)**.** *In the completed Papers 0–IV pipeline, the local Type I alternative is excluded for every finite-energy singular point. Hence every finite-energy singular point belongs to the local Type II alternative in the dichotomy above.*

Proof. This is Theorem 8.5 restated in the language of the local Type I/local Type II dichotomy. □

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